Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
4	(a)		3 protons **and** 4 neutrons (1) 3 leptons **and** 7 baryons (1) 10 ‘up’ **and** 11 ‘down’ quarks (1)	3			3		
	(b)	(i)	Quark-antiquark combination	1			1		
		(ii)	$(+\frac{2}{3} - \frac{2}{3})$ **or** $(+\frac{1}{3} - \frac{1}{3}) = 0$ seen **or** charge of particle **and** its antiparticle is 0		1		1		
		(iii)	udd → udd → $\bar{u}$d uud 4 × 1 mark for each box	4			4		
		(iv)	Production: $\bar{u}$u (or 0) + u = u (1) Decay: u = $\bar{u}$ + uu (1) **Alternative:** Production: -1+1 +1 = [+]1 (1) Decay: 1 = -1+1+1 (1)		2		2		
	(c)		**Required:** • [Very] short decay time (1) **Any × (1):** • Individual quark flavours (numbers) conserved • No neutrino and no gamma emissions • Only {quarks / hadrons} involved	2			2		
			Question 4 total	**10**	**3**	**0**	**13**	**0**	**0**